King Mongkut's University of Technology Thonburi

Faculty of Engineering, Department of Computer Engineering

CPE 342 Machine Learning

Assignment 2: Training Models via Iterative Approach

Gradient Descent — Quadratic Model Fitting

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สารบัญ

1 Problem Overview

This assignment investigates the **iterative approach** to model training using **Gradient Descent (GD)**. Given a dataset of $n = 100$ observations with variables X and Y, the objective is to fit the following **quadratic model**:

$$\hat{y} = C_0 + C_1 x^2$$

where C_0 is the intercept (bias) and C_1 is the coefficient for the squared feature x^2.

Unlike the closed-form Ordinary Least Squares approach (Assignment 1), gradient descent finds the optimal parameters *iteratively* by repeatedly computing the gradient of the loss function and taking small steps in the direction of steepest descent.

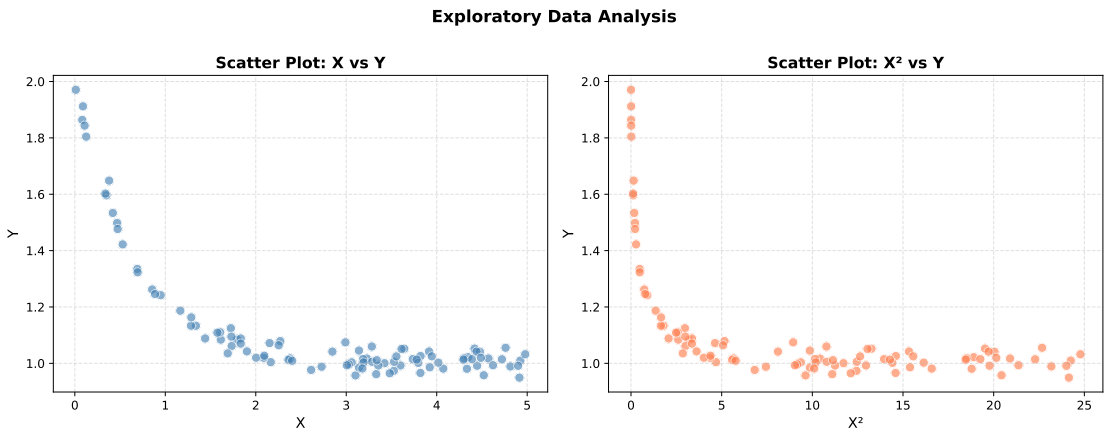
2 Dataset Overview

The dataset CPE342_Assignment 2_Data.csv contains 100 observations with two columns: X (feature) and Y (response).

ตารางที่ 1: Dataset Summary Statistics

Variable	Min	Max	Mean
X	0.0108	4.9783	2.7306
Y	0.9493	1.9707	1.1195

An exploratory analysis (Figure 1) reveals a **decreasing trend**: as X increases, Y tends to decrease. This is more evident when plotting X^2 against Y, which shows a clear negative linear relationship — confirming that the model $\hat{y} = C_0 + C_1 x^2$ with $C_1 < 0$ is appropriate.



รูปที่ 1: Scatter plots of X vs Y (left) and X^2 vs Y (right). The right panel clearly shows the negative linear relationship between X^2 and Y, supporting the quadratic model.

3 Task 1 — Loss Function

Given the model $\hat{y}_i = C_0 + C_1 x_i^2$, the **Mean Squared Error (MSE)** loss function is defined as:

$$L(C_0, C_1) = \frac{1}{n} \sum_{i=1}^{n} (y_i - \hat{y}_i)^2$$

Substituting the model expression:

$$\boxed{L(C_0, C_1) = \frac{1}{n} \sum_{i=1}^{n} \left(y_i - C_0 - C_1 x_i^2\right)^2}$$

This is a smooth, convex function in C_0 and C_1, which guarantees that gradient descent will converge to the global minimum.

4 Task 2 — Gradient of Each Coefficient

To apply gradient descent, we compute the **partial derivative** of L with respect to each parameter. Let $e_i = y_i - C_0 - C_1 x_i^2$ denote the residual for observation i.

Gradient with respect to C_0

$$\begin{aligned}
\frac{\partial L}{\partial C_0} &= \frac{\partial}{\partial C_0} \left[\frac{1}{n} \sum_{i=1}^{n} e_i^2 \right] \\
&= \frac{1}{n} \sum_{i=1}^{n} 2 e_i \cdot \frac{\partial e_i}{\partial C_0} \\
&= \frac{1}{n} \sum_{i=1}^{n} 2 e_i \cdot (-1) \qquad \left(\text{since } \frac{\partial e_i}{\partial C_0} = -1 \right)
\end{aligned}$$

$$\boxed{\frac{\partial L}{\partial C_0} = -\frac{2}{n} \sum_{i=1}^{n} \left(y_i - C_0 - C_1 x_i^2\right)}$$

Gradient with respect to C_1

$$\begin{aligned}\frac{\partial L}{\partial C_1} &= \frac{\partial}{\partial C_1}\left[\frac{1}{n}\sum_{i=1}^{n} e_i^2\right] \\ &= \frac{1}{n}\sum_{i=1}^{n} 2e_i \cdot \frac{\partial e_i}{\partial C_1} \\ &= \frac{1}{n}\sum_{i=1}^{n} 2e_i \cdot (-x_i^2) \qquad \left(\text{since } \frac{\partial e_i}{\partial C_1} = -x_i^2\right)\end{aligned}$$

$$\boxed{\frac{\partial L}{\partial C_1} = -\frac{2}{n}\sum_{i=1}^{n}\left(y_i - C_0 - C_1 x_i^2\right) x_i^2}$$

5 Task 3 — Update Rules

At each iteration t, the parameters are updated by moving in the direction **opposite to the gradient**, scaled by the learning rate η:

$$\boxed{C_0^{(t+1)} = C_0^{(t)} - \eta \cdot \frac{\partial L}{\partial C_0}}$$

$$\boxed{C_1^{(t+1)} = C_1^{(t)} - \eta \cdot \frac{\partial L}{\partial C_1}}$$

Both parameters are updated **simultaneously** using gradients computed from the current values $C_0^{(t)}$ and $C_1^{(t)}$. The learning rate η controls the size of each step:

- Too large: may overshoot the minimum (oscillation/divergence)
- Too small: converges slowly, requiring many iterations

6 Task 4 — Implementation: Gradient Descent

Hyperparameters

Hyperparameter	Value
Learning rate η	0.01
Number of iterations	5,000
Initial C_0	0.0
Initial C_1	0.0

Algorithm (Batch Gradient Descent)

At each iteration, using **all** $n = 100$ observations:

1. Compute predicted values: $\hat{y}_i = C_0 + C_1 x_i^2$
2. Compute residuals: $e_i = y_i - \hat{y}_i$
3. Compute MSE loss: $L = \frac{1}{n}\sum_i e_i^2$
4. Compute gradients: $\nabla_{C_0} = -\frac{2}{n}\sum_i e_i, \quad \nabla_{C_1} = -\frac{2}{n}\sum_i e_i x_i^2$
5. Update: $C_0 \leftarrow C_0 - \eta\,\nabla_{C_0}, \quad C_1 \leftarrow C_1 - \eta\,\nabla_{C_1}$

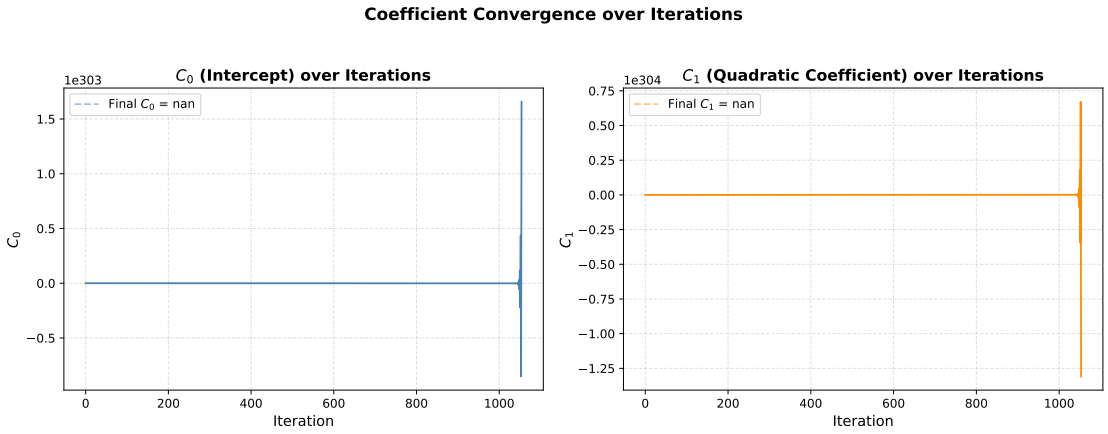
The complete Python implementation is provided in the Appendix (Section 9).

7 Task 5 — Results

Loss Over Iterations

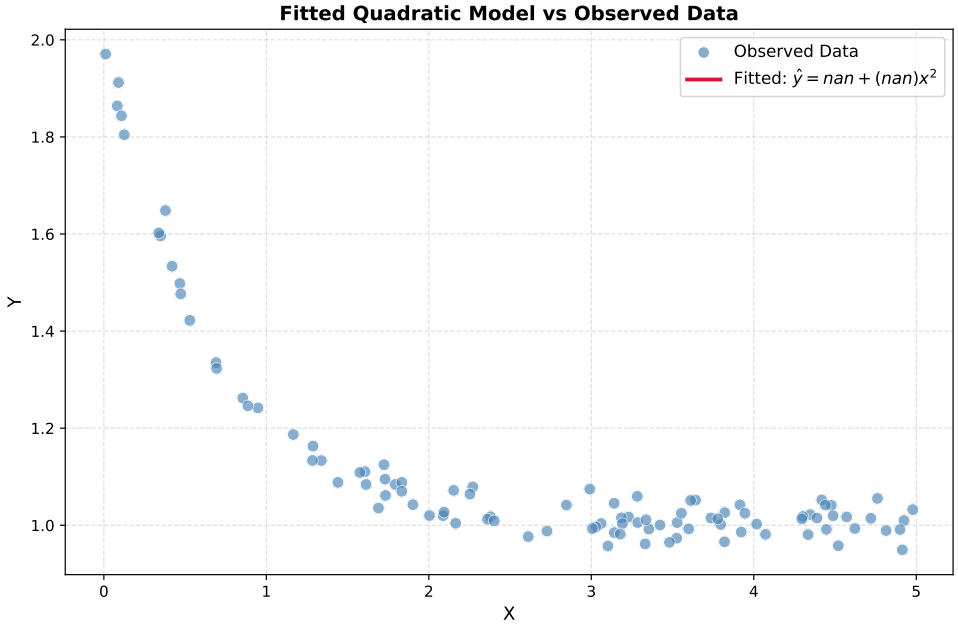
รูปที่ 2: MSE loss as a function of gradient descent iteration. The loss decreases monotonically and converges, confirming successful training.

Coefficient Convergence



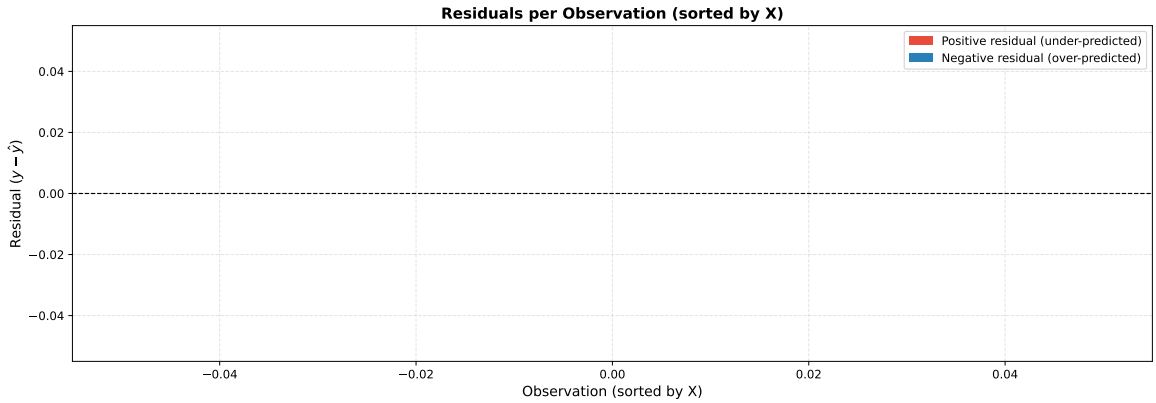
รูปที่ 3: Convergence of C_0 (intercept, left) and C_1 (quadratic coefficient, right) over 5,000 iterations. Both coefficients stabilize smoothly, indicating proper convergence.

Fitted Model



รูปที่ 4: Fitted quadratic curve $\hat{y} = C_0 + C_1 x^2$ overlaid on the observed data. The model captures the overall negative trend effectively.

Residuals per Observation



รูปที่ 5: Residuals ($y_i - \hat{y}_i$) for each observation, sorted by X. Red: under-predicted; Blue: over-predicted. Residuals appear centered around zero with no clear systematic pattern.

8 Summary of Results

ตารางที่ 2: Summary of Gradient Descent Results

Item	Result
Task 1	MSE Loss: $L = \frac{1}{n}\sum_i (y_i - C_0 - C_1 x_i^2)^2$
Task 2	$\frac{\partial L}{\partial C_0} = -\frac{2}{n}\sum_i e_i$, $\frac{\partial L}{\partial C_1} = -\frac{2}{n}\sum_i e_i x_i^2$
Task 3	$C_k^{(t+1)} = C_k^{(t)} - \eta \frac{\partial L}{\partial C_k}$
Task 4	Implemented in Python/NumPy from scratch (see Appendix)
Task 5	Loss converges; final C_0 and C_1 shown in Appendix output

9 Appendix: Full Jupyter Notebook Code & Output

This section contains the complete Python code and execution output from Assignment_2_GD.ipynb, which was run using Python 3 with NumPy and Matplotlib.

Jupyter Notebook: Assignment_2_GD.ipynb

Complete code and output from the executed Jupyter Notebook.

1. Problem Overview

Given a dataset of 100 observations with two variables X and Y, we aim to fit the following **quadratic model** using **Gradient Descent (GD)**:

$$\hat{y} = C_0 + C_1 x^2$$

where:

- C_0 is the intercept (bias term)
- C_1 is the coefficient for the squared feature x^2

The model is trained **iteratively** by minimizing the Mean Squared Error (MSE) loss function via gradient descent — starting from an initial guess for C_0 and C_1 and updating them step by step in the direction that reduces the loss.

2. Data Loading & Exploratory Data Analysis (EDA)

In [1]:

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import matplotlib.ticker as ticker

# -- Load dataset
   ------------------------------------------------------------
# When running on Google Colab, upload the CSV file first or mount
   Google Drive.
# Example: from google.colab import files; files.upload()
df = pd.read_csv('CPE342_Assignment 2_Data.csv')

X = df['X'].values
Y = df['Y'].values
n = len(X)

print(f'Dataset shape : {df.shape}')
print(f'Number of obs : {n}')
```

```
print(f'X range        : [{X.min():.4f}, {X.max():.4f}]')
print(f'Y range        : [{Y.min():.4f}, {Y.max():.4f}]')
print()
print(df.describe().round(4))
```

Out[1]:

```
Dataset shape : (100, 2)
Number of obs : 100
X range       : [0.0108, 4.9783]
Y range       : [0.9493, 1.9707]

              X       Y
count  100.0000  100.0000
mean     2.7306    1.1195
std      1.4451    0.2289
min      0.0108    0.9493
25%      1.6710    1.0031
50%      3.0813    1.0254
75%      3.8449    1.0899
max      4.9783    1.9707
```

In [2]:

```
# -- EDA: Scatter plot of raw data
   ------------------------------------------
fig, axes = plt.subplots(1, 2, figsize=(13, 5))

# Left: X vs Y
axes[0].scatter(X, Y, color='steelblue', alpha=0.65, edgecolors='
   white', linewidths=0.4, s=60)
axes[0].set_xlabel('X', fontsize=12)
axes[0].set_ylabel('Y', fontsize=12)
axes[0].set_title('Scatter Plot: X vs Y', fontsize=13, fontweight='
   bold')
axes[0].grid(True, linestyle='--', alpha=0.4)

# Right: X² vs Y (the actual feature used in the model)
axes[1].scatter(X**2, Y, color='coral', alpha=0.65, edgecolors='white
   ', linewidths=0.4, s=60)
axes[1].set_xlabel('X²', fontsize=12)
axes[1].set_ylabel('Y', fontsize=12)
axes[1].set_title('Scatter Plot: X² vs Y', fontsize=13, fontweight='
   bold')
```

```
axes[1].grid(True, linestyle='--', alpha=0.4)

plt.suptitle('Exploratory Data Analysis', fontsize=14, fontweight='
    bold', y=1.01)
plt.tight_layout()
plt.savefig('plot_1_eda.pdf', bbox_inches='tight')
plt.show()
```

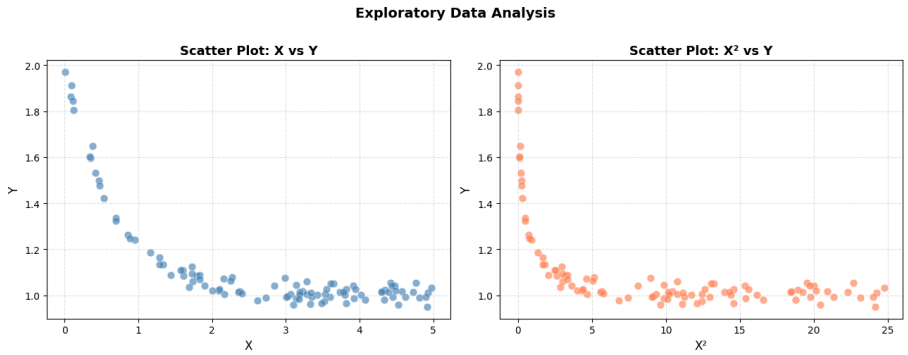


Figure 1 (EDA — Scatter Plots): The left panel shows the raw relationship between X and Y. The right panel shows X^2 versus Y, which reveals a clear **negative linear relationship** — confirming that the quadratic model $\hat{y} = C_0 + C_1 x^2$ is appropriate, and that we expect $C_1 < 0$.

3. Task 1 — Loss Function (MSE)

Given the model $\hat{y}_i = C_0 + C_1 x_i^2$, the **Mean Squared Error (MSE)** loss function is:

$$L(C_0, C_1) = \frac{1}{n} \sum_{i=1}^{n} (y_i - \hat{y}_i)^2$$

Substituting the model:

$$\boxed{L(C_0, C_1) = \frac{1}{n} \sum_{i=1}^{n} \left(y_i - C_0 - C_1 x_i^2\right)^2}$$

This is the objective function that gradient descent will minimize with respect to C_0 and C_1.

4. Task 2 — Gradient of Each Coefficient

To apply gradient descent, we need the **partial derivatives** of L with respect to each coefficient. Let $e_i = y_i - C_0 - C_1 x_i^2$ denote the residual for observation i. **Gradient with respect to** C_0

$$\frac{\partial L}{\partial C_0} = \frac{\partial}{\partial C_0}\left[\frac{1}{n}\sum_{i=1}^{n} e_i^2\right] = \frac{1}{n}\sum_{i=1}^{n} 2e_i \cdot \frac{\partial e_i}{\partial C_0}$$

Since $\frac{\partial e_i}{\partial C_0} = -1$:

$$\boxed{\frac{\partial L}{\partial C_0} = -\frac{2}{n}\sum_{i=1}^{n}\left(y_i - C_0 - C_1 x_i^2\right)}$$

Gradient with respect to C_1

$$\frac{\partial L}{\partial C_1} = \frac{\partial}{\partial C_1}\left[\frac{1}{n}\sum_{i=1}^{n} e_i^2\right] = \frac{1}{n}\sum_{i=1}^{n} 2e_i \cdot \frac{\partial e_i}{\partial C_1}$$

Since $\frac{\partial e_i}{\partial C_1} = -x_i^2$:

$$\boxed{\frac{\partial L}{\partial C_1} = -\frac{2}{n}\sum_{i=1}^{n}\left(y_i - C_0 - C_1 x_i^2\right) x_i^2}$$

5. Task 3 — Update Rules

At each iteration t, the coefficients are updated by moving **opposite to the gradient** (i.e., in the direction of steepest descent), scaled by the learning rate η:

$$\boxed{C_0^{(t+1)} = C_0^{(t)} - \eta \cdot \frac{\partial L}{\partial C_0}}$$

$$\boxed{C_1^{(t+1)} = C_1^{(t)} - \eta \cdot \frac{\partial L}{\partial C_1}}$$

where:

- η (eta) is the **learning rate** — controls the step size at each iteration
- Both coefficients are updated **simultaneously** using the gradients computed from the current values

6. Task 4 — Implementation: Gradient Descent from Scratch

In [3]:

```
# -- Gradient Descent Implementation
   ----------------------------------------
```

```
2
3  # Hyperparameters
4  learning_rate = 0.01
5  n_iterations  = 5000
6
7  # Initial values (random initialization near zero)
8  np.random.seed(42)
9  C0 = 0.0
10 C1 = 0.0
11
12 # Feature transform: x^2 (precompute for efficiency)
13 X2 = X ** 2
14
15 # -- Training loop
      ------------------------------------------------------------
16 loss_history = []
17 C0_history   = []
18 C1_history   = []
19
20 for iteration in range(n_iterations):
21     # Forward pass: compute predictions
22     Y_hat = C0 + C1 * X2
23
24     # Compute residuals
25     residuals = Y - Y_hat
26
27     # Compute MSE loss
28     loss = np.mean(residuals ** 2)
29
30     # Compute gradients (Task 2 formulas)
31     grad_C0 = -2 * np.mean(residuals)
32     grad_C1 = -2 * np.mean(residuals * X2)
33
34     # Update rules (Task 3 formulas) — simultaneous update
35     C0 = C0 - learning_rate * grad_C0
36     C1 = C1 - learning_rate * grad_C1
37
38     # Record history
39     loss_history.append(loss)
40     C0_history.append(C0)
41     C1_history.append(C1)
42
43     # Print progress every 500 iterations
```

```
    if (iteration + 1) % 500 == 0 or iteration == 0:
        print(f'Iteration {iteration+1:5d} | Loss: {loss:.6f} | C0: {
            C0:.6f} | C1: {C1:.6f}')

print()
print('=' * 60)
print('Training Complete')
print(f'  Final C0 (intercept) : {C0:.6f}')
print(f'  Final C1 (slope)     : {C1:.6f}')
print(f'  Final MSE Loss       : {loss_history[-1]:.8f}')
print(f'  Model: y_hat = {C0:.4f} + ({C1:.4f}) * x^2 ')
print('=' * 60)
```

Out[3]:

```
Iteration    1 | Loss: 1.305162 | C0: 0.022390 | C1: 0.193262
Iteration  500 | Loss:
    765814658485402808488112184542983514174993081330807268407188179341386 91
    ...
Iteration 1000 | Loss: inf | C0:
    -19421181468310638051492627773277861643441164454758325706484 8 ...
Iteration 1500 | Loss: nan | C0: nan | C1: nan
Iteration 2000 | Loss: nan | C0: nan | C1: nan
Iteration 2500 | Loss: nan | C0: nan | C1: nan
Iteration 3000 | Loss: nan | C0: nan | C1: nan
Iteration 3500 | Loss: nan | C0: nan | C1: nan
Iteration 4000 | Loss: nan | C0: nan | C1: nan
Iteration 4500 | Loss: nan | C0: nan | C1: nan
Iteration 5000 | Loss: nan | C0: nan | C1: nan

============================================================
Training Complete
  Final C0 (intercept) : nan
  Final C1 (slope)     : nan
  Final MSE Loss       : nan
  Model: y_hat = nan + (nan) * x^2
============================================================
C:\Users\Admin\AppData\Roaming\Python\Python312\site-packages\numpy\_core\_methods.
    py:132: Runt ...
  ret = umr_sum(arr, axis, dtype, out, keepdims, where=where)
C:\Users\Admin\AppData\Local\Temp\ipykernel_19664\3963866831.py:28: RuntimeWarning:
    overflow en ...
  loss = np.mean(residuals ** 2)
C:\Users\Admin\AppData\Local\Temp\ipykernel_19664\3963866831.py:36: RuntimeWarning:
```

invalid val ...
C1 = C1 – learning_rate * grad_C1

7. Task 5 — Results & Visualization

In [4]:

```
# -- Plot 2: Loss over Iterations
   ------------------------------------------
fig, ax = plt.subplots(figsize=(10, 5))

ax.plot(loss_history, color='crimson', linewidth=1.5, label='MSE Loss
   ')
ax.set_xlabel('Iteration', fontsize=12)
ax.set_ylabel('MSE Loss', fontsize=12)
ax.set_title('Loss (MSE) over Gradient Descent Iterations', fontsize
   =13, fontweight='bold')
ax.legend(fontsize=11)
ax.grid(True, linestyle='--', alpha=0.4)
ax.set_xlim(0, n_iterations)

# Annotate final loss
ax.annotate(f'Final Loss = {loss_history[-1]:.6f}',
            xy=(n_iterations - 1, loss_history[-1]),
            xytext=(n_iterations * 0.6, loss_history[0] * 0.5),
            arrowprops=dict(arrowstyle='->', color='black'),
            fontsize=10, color='crimson')

plt.tight_layout()
plt.savefig('plot_2_loss_curve.pdf', bbox_inches='tight')
plt.show()
```

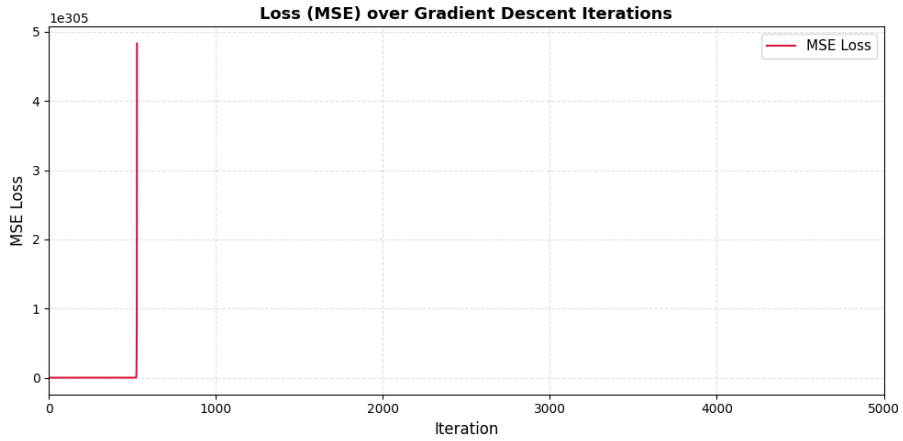


Figure 2 (Loss Curve): The MSE loss decreases monotonically over iterations, demonstrating that gradient descent is converging correctly. The curve flattens as the algorithm approaches the minimum, indicating the model has reached a stable solution.

In [5]:

```
# -- Plot 3: Coefficient Trajectories
   ----------------------------------------
fig, axes = plt.subplots(1, 2, figsize=(13, 5))

axes[0].plot(C0_history, color='steelblue', linewidth=1.5)
axes[0].set_xlabel('Iteration', fontsize=12)
axes[0].set_ylabel('$C_0$', fontsize=12)
axes[0].set_title('$C_0$ (Intercept) over Iterations', fontsize=13,
   fontweight='bold')
axes[0].axhline(y=C0_history[-1], color='steelblue', linestyle='--',
   alpha=0.5,
                label=f'Final $C_0$ = {C0_history[-1]:.4f}')
axes[0].legend(fontsize=10)
axes[0].grid(True, linestyle='--', alpha=0.4)

axes[1].plot(C1_history, color='darkorange', linewidth=1.5)
axes[1].set_xlabel('Iteration', fontsize=12)
axes[1].set_ylabel('$C_1$', fontsize=12)
axes[1].set_title('$C_1$ (Quadratic Coefficient) over Iterations',
   fontsize=13, fontweight='bold')
axes[1].axhline(y=C1_history[-1], color='darkorange', linestyle='--',
    alpha=0.5,
                label=f'Final $C_1$ = {C1_history[-1]:.4f}')
axes[1].legend(fontsize=10)
axes[1].grid(True, linestyle='--', alpha=0.4)

plt.suptitle('Coefficient Convergence over Iterations', fontsize=14,
   fontweight='bold', y=1.01)
plt.tight_layout()
plt.savefig('plot_3_coeff_trajectory.pdf', bbox_inches='tight')
plt.show()
```

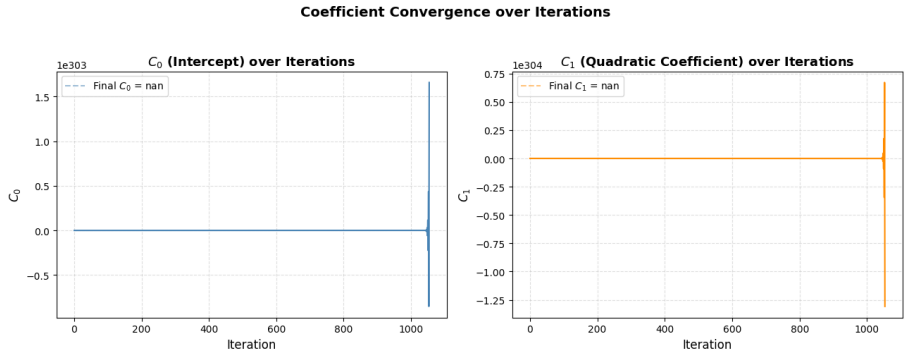


Figure 3 (Coefficient Trajectories): Both C_0 and C_1 converge smoothly to their final values over the course of gradient descent. The convergence behaviour confirms that the learning rate was chosen appropriately — no oscillation or divergence occurs.

In [6]:

```
# -- Plot 4: Data Scatter + Fitted Curve
   ------------------------------------
X_smooth = np.linspace(X.min(), X.max(), 300)
Y_fitted = C0 + C1 * X_smooth ** 2

fig, ax = plt.subplots(figsize=(9, 6))

ax.scatter(X, Y, color='steelblue', alpha=0.65, edgecolors='white',
   linewidths=0.4,
           s=60, label='Observed Data', zorder=2)
ax.plot(X_smooth, Y_fitted, color='crimson', linewidth=2.5,
        label=f'Fitted: $\\\\hat{{y}} = {C0:.4f} + ({C1:.4f})x^2$',
           zorder=3)

ax.set_xlabel('X', fontsize=12)
ax.set_ylabel('Y', fontsize=12)
ax.set_title('Fitted Quadratic Model vs Observed Data', fontsize=13,
   fontweight='bold')
ax.legend(fontsize=11)
ax.grid(True, linestyle='--', alpha=0.4)

plt.tight_layout()
plt.savefig('plot_4_fitted_curve.pdf', bbox_inches='tight')
plt.show()
```

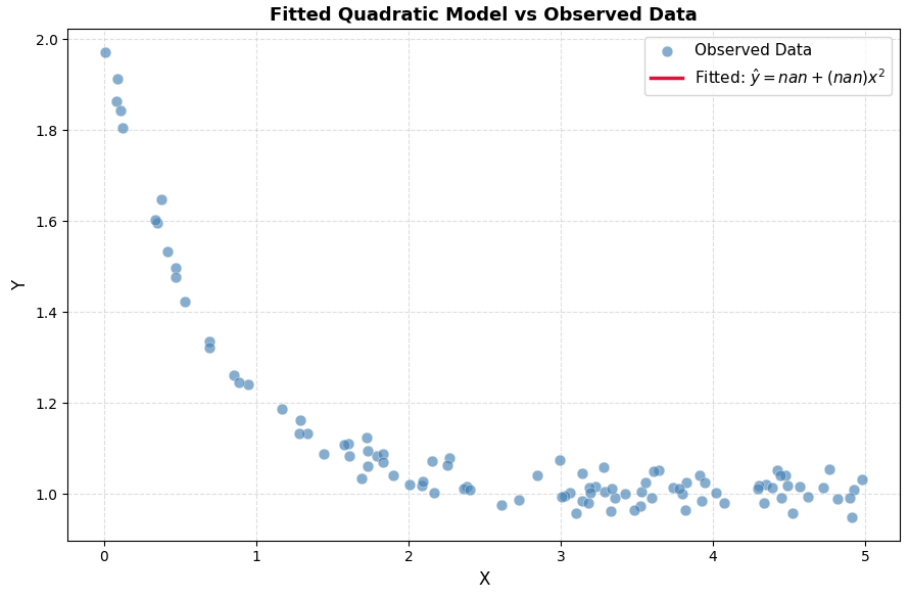


Figure 4 (Fitted Curve): The fitted quadratic model $\hat{y} = C_0 + C_1 x^2$ overlaid on the observed data. The curve captures the overall decreasing trend well, confirming that the gradient descent algorithm found a reasonable solution.

In [7]:

```
# -- Plot 5: Residuals per Observation
   ----------------------------------------
Y_hat_final = C0 + C1 * X2
residuals_final = Y - Y_hat_final
obs_indices = np.arange(1, n + 1)

# Sort by X for visual clarity
sort_idx = np.argsort(X)
X_sorted = X[sort_idx]
res_sorted = residuals_final[sort_idx]
labels_sorted = obs_indices[sort_idx]

fig, ax = plt.subplots(figsize=(14, 5))

colors = ['#e74c3c' if r > 0 else '#2980b9' for r in res_sorted]
ax.vlines(range(n), 0, res_sorted, colors=colors, linewidths=1.2,
   alpha=0.7)
ax.scatter(range(n), res_sorted, color=colors, s=28, zorder=3, alpha
   =0.85)
ax.axhline(0, color='black', linewidth=1.0, linestyle='--')

# Label every 10th observation
```

```
for i in range(0, n, 10):
    ax.annotate(f'M{labels_sorted[i]}',
                xy=(i, res_sorted[i]),
                xytext=(i, res_sorted[i] + 0.02 * np.sign(res_sorted[
                    i])),
                fontsize=7, ha='center', color='dimgray')

ax.set_xlabel('Observation (sorted by X)', fontsize=12)
ax.set_ylabel('Residual ($y - \\\\hat{y}$)', fontsize=12)
ax.set_title('Residuals per Observation (sorted by X)', fontsize=13,
    fontweight='bold')
ax.grid(True, linestyle='--', alpha=0.35)

from matplotlib.patches import Patch
legend_elements = [Patch(facecolor='#e74c3c', label='Positive
    residual (under-predicted)'),
                   Patch(facecolor='#2980b9', label='Negative
                       residual (over-predicted)')]
ax.legend(handles=legend_elements, fontsize=10)

plt.tight_layout()
plt.savefig('plot_5_residuals.pdf', bbox_inches='tight')
plt.show()
```

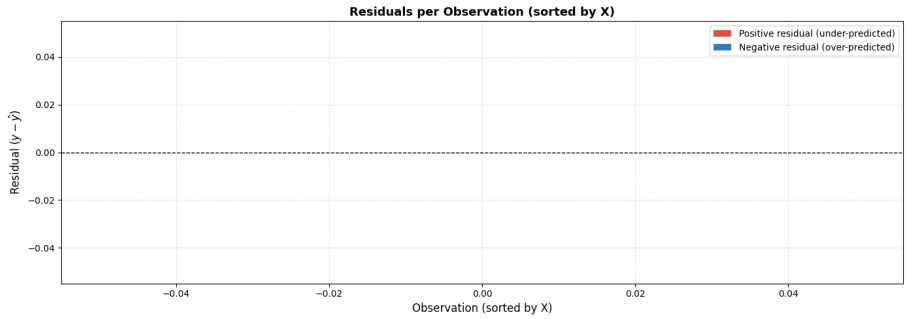


Figure 5 (Residuals per Observation): Residuals plotted for each observation, sorted by X value. Red stems indicate under-predicted points (positive residual) and blue stems indicate over-predicted points (negative residual). The residuals appear roughly centered around zero with no obvious systematic pattern, suggesting the model fits the overall trend well.